LASRMIS: LLM-Assisted Structured Reporting from Medical Image Segmentation – A Prompt Ablation Study on Prostate MRI

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Abstract. We introduce **LASRMIS** (LLM-Assisted Structured Reporting from Medical Image Segmentation), a training-free pipeline that translates prostate MRI segmentation masks into structured radiology reports by feeding three varying complexity prompting strategies to a large language model. We evaluated three prompt strategies of varying complexity (minimal, structured, and full-context) in 32 cases from the Medical Decathlon Task05 dataset and evaluated the generated reports for clinical completeness, faithfulness and readability. A validated ten-element rubric ($\kappa = 0.864$) assesses adherence to PI-RADS v2.1 guidelines. Full-context prompting achieved a completeness score 86.2% vs. 37.5% for minimal prompting, a difference of 48.7 percentage points, all due to the design of the prompt with all differences statistically significant (Wilcoxon, Bonferroni-corrected, $p < 0.001$). Both structured and full-context prompts got zero hallucinations. This work identifies prompt engineering as a major influence on the quality of clinical reports in segmentation-to-report pipelines, without any need for model fine-tuning or specialized infrastructure.

Keywords: Prostate MRI · Generation of radiology reports · Large Language Models · Prompt engineering · Medical image segmentation · PI-RADS

1 Introduction

Prostate cancer is one of the most commonly diagnosed cancers in men globally, and multi-parametric MRI (mpMRI) has emerged as the preferred imaging technique for its detection and staging. Once the scan is completed, radiologists should convert quantitative segmentation results into refined written reports, which can be labor-intensive and a source of workflow bottlenecks in busy imaging centers. Automating this step using deep learning has attracted growing research interest in recent years [1,2].

LLMs have shown real promise across a range of radiology tasks [3], from drafting reports to extracting clinical findings and supporting diagnostic decisions. What remains surprisingly underexplored, though is how much the design of the prompt itself affects the quality of the output—especially when reports are generated from segmentation metrics rather than from images or existing text. Prior work on prompt engineering for medical question answering has shown that prompt strategy can meaningfully shift LLM outputs [4,5,6], but those insights have not yet been applied to the specific problem of report generation from segmentation data.

Most existing segmentation-to-report approaches work directly with raw image features rather than structured metrics [7], or go in the opposite direction altogether—extracting information from reports that already exist [8]. Vision-language models [9,10] and multimodal report generation systems [11,12] have pushed the field forward considerably, yet pipelines that need no training at all and operate purely on segmentation metric outputs have received little systematic attention.

This paper makes three contributions. First, **LASRMIS** is a fully training-free pipeline that converts prostate MRI segmentation masks into structured radiology reports using LLM prompting, with no model fine-tuning or dedicated infrastructure required. Second, to the best of our knowledge, we carried out the first systematic prompt ablation study that directly measures how prompt complexity shapes report completeness, faithfulness and readability in the segmentation-metric-to-report setting. Third, we introduce a validated ten-element clinical completeness rubric for prostate MRI report assessment ($\kappa = 0.864$) that can be reused in future work.

2 Related Work

2.1 Automated Medical Report Generation

Automated medical report generation has been investigated via deep learning models such as retrieval-based methods, attention mechanisms, reinforcement learning, and LLM-based architectures [1,2]. These surveys pointed out that information fusion from multiple modalities and better clinical evaluation metrics remain key open challenges. An orthogonal line is segmentation-driven report generation, which translates structured outputs of segmentation models to clinical narratives [7]. LASRMIS makes progress in this direction by mapping segmentation metrics into structured reports without any model training.

2.2 LLMs in Radiology

LLMs have been applied to various radiology tasks [3] with a scoping review of 69 studies documenting that structured prompting and fine-tuning can improve performance. Radiology impressions have been generated from GPT-4o using in-context learning [11], with semantic nearest-neighbor methods as the top-performing approaches on BERTScore, ROUGE and METEOR. Multimodal LLMs have been tested on CT scan interpretation using fully automated evaluation methods [12]. Encoder-based models versus generative LLMs have been compared for named entity recognition in clinical reports [13], showing that the task type significantly affects the best model class. It has also been shown that LLM performance in radiology varies over time, and that there is a need for continuous benchmarking [14]. LASRMIS is motivated by the inherently different problem of report generation from segmentation results instead of from images or texts.

2.3 Vision-Language Models for Medical Imaging

VLMs have advanced quickly across a wide range of medical imaging tasks—from classification and segmentation to report generation and visual question answering [9,10]. An interesting example is CBVLM [15], a concept-based framework that requires no training yet achieves competitive results by grounding predictions in clinically meaningful visual concepts. LASRMIS is also training-free but it applies in a text domain only—that is to say it transforms structured numeric measurements into human readable clinical narratives rather than processing images.

2.4 Prompt Engineering for Clinical AI

The phrasing of the prompt is very important for the LLM when it comes to medical document. When Chain-of-Thought prompting was also tested on 5 LLMs on clinical datasets [4], the model comparing questions and answers model architecture is more important than the prompt style for question-answering task – We extend this observation to report generation completeness in terms of a complexity of prompt only significantly increases it by a 48.7% point gap. A cheat sheet for radiology profession prompting [6] even highlights that you need to have a well-written prompt to get a precise, context aware radiology report. Work on medical question answering [5] also suggests that the selection of the appropriate prompting method significantly impacts performance, the ReAct style prompting attaining up to 81.7% accuracy. Hybrid Chain of Thought reasoning with retrieval-augmented generation, and fine-tuning has also been found to have considerable potential for boosting clinical decision support trustworthiness [16]. Multimodal prompting for AD report generation [17] further corroborates that prompt engineering carefully outperforms generalization for clinical domains.

2.5 Medical AI: Trustworthiness and Safety

The use of LLMs in medical practice raises important questions of trust and safety. The multi-layered architecture of a full trustworthy AI system [18] includes robustness and transparency to the design and human in the loop. Metamorphic testing [19] has also been used to evaluate robustness of LLMs in medicine, stating that minor input perturbations may lead to drastic changes in outputs of clinical models. LASRMIS Prompts B and C achieved almost no hallucination, thus alleviating these security concerns.

3 Methodology

3.1 Dataset

We used the Medical Segmentation Decathlon Task05 Prostate dataset, which consists of 32 prostate MRI cases with expert ground-truth segmentation masks. The T2-weighted (T2W) and apparent diffusion coefficient (ADC) modalities are available for each case. The segmentation labels for peripheral zone (label 1) and transition zone (label 2) are included. This publicly available benchmark allows for full reproducibility.

3.2 Metric Extraction

A metric extractor then derives structured clinical features from each segmentation mask, which may be passed directly to a prompt. To guaranty correct volume computation, the voxel spacing is obtained from the NIfTI file header. From each zone, individual lesion regions are detected by connected component labeling (`scipy.ndimage.label`). For all region, we record the volume in millilitre, the anatomical zone together with a PI-RADS proxy score that is derived from volume threshold taken from PI-RADS v2.1 size criteria. Peripheral zone lesions smaller than 0.5 mL are assigned the PI-RADS score of 3; 0.5–1.5 mL, PI-RADS 4; and larger than 1.5 mL, PI-RADS 5. Transition zone thresholds are derived in the same way. Regions smaller than 0.1 mL are eliminated as segmentation noise. However, one should note that these PI-RADS are only proxies—Complete PI-RADS v2.1 scoring is based on T2 signal, ADC maps, dynamic contrast enhancement, of which none are directly obtainable from segmentation masks.

3.3 LLM Inference

All of the outputs were generated using LLaMA 3.1-8B-Instant through the Groq API. Temperature was set to 0 at all times to make output deterministic and reproducible, and `max_tokens=500`. All 32 cases were executed on all three prompt types, yielding a total of 96 reports. To cope with sporadic API timeout, a retry mechanism was incorporated with a maximum of 3 attempts and a wait of 3 seconds between each.

3.4 Prompt Strategies

Three prompt strategies of increasing complexity represent a natural progression from unguided to best-practice clinical prompt engineering, motivated by prior work on prompt design in medical AI [6,5] (Table 1).

Table 1. Summary of the three prompt strategies evaluated.

Strategy	Key Elements	Description
Prompt A (Minimal)	Raw numbers only	Three numeric fields (regions, volume, PI-RADS) with a single unstructured instruction "Explain this." No role, no format.
Prompt B (Structured)	Role + organised input + 4 output sections	Assigns medical AI role, labels input fields, specifies four output sections: key finding, location/volume, clinical significance, recommended next step.
Prompt C (Full-Context)	All of B + disclaimer + per-region PI-RADS + uncertainty flags + 5 section schema + language instructions	Adds mandatory radiologist review disclaimer, per-region PI-RADS estimates, uncertainty flags conditioned on PI-RADS level, five explicit section headings, and a 2–3 sentence per section constraint.

3.5 Evaluation Framework

Clinical Completeness A ten-element rubric was created based on PI-RADS v2.1 structured reporting protocols. Each report is scored for the presence of: lesion identification, volumetric measurement, zone specification, PI-RADS estimate, biopsy recommendation, follow-up plan, clinical significance assessment, uncertainty disclaimer, peripheral zone reference and transition zone reference. Score is the proportion of elements found (0–100%).

Faithfulness A set of 20 clinically realistic but input-unsupported terms was constructed based on frequent prostate MRI descriptors not part of the segmentation metric set (e.g., diffusion restriction, irregular margins, capsular contact). Each report was subjected to automatic screening and assigned a faithfulness score of 3 (zero unsupported terms), 2 (one term), 1 (two terms), or 0 (three or more terms). Hallucination in LLM-generated clinical text is a central safety issue across the literature [3,18].

Readability Flesch Reading Ease was calculated via the `textstat` Python module. For clinical reports, lower scores reflect appropriate medical register. Readability has been recognized as an important evaluation factor in LLM-based radiology report assessment [3].

Inter-Rater Reliability The reliability of the rubric was evaluated by testing the level of agreement between a conservative rater (core clinical keywords) and a liberal rater (broader lexicon of synonyms) on a stratified sample of 30 reports (10 per prompt strategy, 31.3%. Cohen's kappa was $\kappa = 0.864$ (Almost Perfect), with kappa per-prompt ranging from 0.787 to 0.860. Three features (lesion, volume, PI-RADS) were consistently present, resulting in perfect agreement by definition; the other seven can be viewed as random variables with a mean $\kappa = 0.718$ (Substantial), supporting the stability of the rubric.

Statistical Analysis Pairwise completeness discrepancies were assessed by the Wilcoxon signed-rank test with a Bonferroni correction ($\alpha = 0.05/3 = 0.017$). Effect size was calculated as $r = |Z|/\sqrt{N}$. The own-subject design controls for case-level variance between the three conditions.

4 Experimental Results

4.1 Pipeline Overview

Figure 1 shows the entire LASRMIS pipeline starting from MRI input, ranging over metric extraction, prompt construction, LLM inference, multi-dimensional evaluation, and final results.

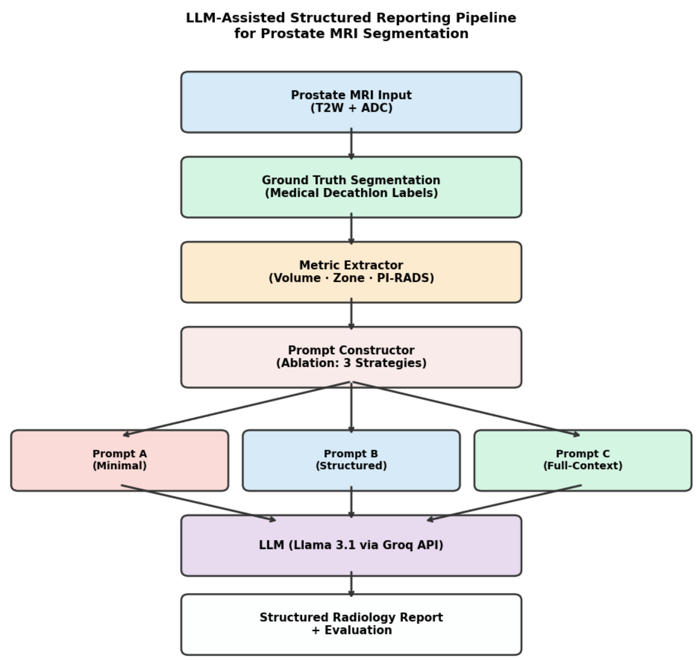

Fig. 1. LASRMIS pipeline.The segmentation masks are transformed into structured metrics, which are organized into three line strategies of escalating complexity, sent to LLaMA 3.1 through Groq API, and assessed on completeness, faithfulness and readability.

4.2 Quantitative Evaluation

Table 2 reports the full results of our evaluation. Figure 2 shows all the three dimensions in the form of bar charts with error bars representing standard deviation.

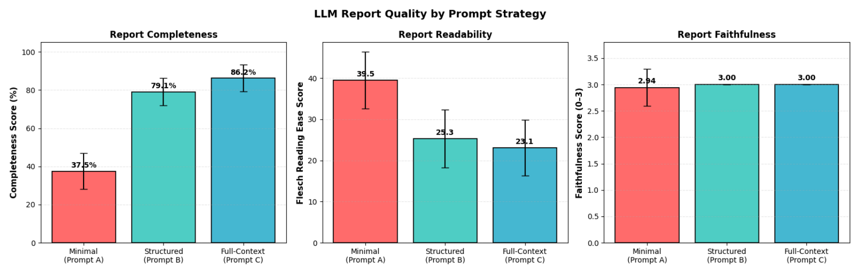

Fig. 2. A quantitative assessment of report quality under the three prompt strategies. Left: Clinical completeness (%) demonstrates a 48.7 pp difference between Minimal (A) and Full-Context (C) prompting. Center: Flesch Reading Ease scores show that structured prompts lead to a more suitable, professional clinical register (lower scores). Right: Fidelity scores (0–3) indicate that increased prompt complexity eliminates hallucinations, achieving maximum groundedness with the Structured and Full-Context groups. Error bars show standard deviation for the 32 cases.

Table 2. Quality of reports among three prompt strategies: a multi-dimensional evaluation (32 cases, 96 reports). Clinical Completeness Clinical Completeness is graded with a ten item rubric (0–100%), and Fidelity is rated on a 0–3 (3 = strictly grounded) scale. Flesch Reading Ease scores show clinical register, with lower scores indicating more appropriate professional language. The Hallucination rate is the percentage of reports with at least an unsupported term. The results show that as prompt complexity increases, completeness monotonically improves and hallucinations are entirely suppressed for both the Structured and Full-Context approaches.

Strategy	**Completeness (%)**	**Faithfulness**	**Flesch**	**Hall. Rate**
Minimal (A)	37.5 ± 9.4	2.94 ± 0.35	39.5 ± 6.9	3.1%
Structured (B)	79.1 ± 7.2	3.00 ± 0.00	25.3 ± 7.0	0.0%
Full-Context (C)	86.2 ± 7.0	3.00 ± 0.00	23.1 ± 6.8	0.0%

4.3 Statistical Significance

Table 3 presents the results of the Wilcoxon signed-rank test. Each test was significant with a large effect size after Bonferroni correction. The W = 0 statistic for A-vs-B and for A-vs-C means that all 32 cases improved monotonically – zero exceptions.

Table 3. Wilcoxon signed-rank test results with Bonferroni correction ($\alpha_{\text{corrected}} = 0.017$) for pairwise comparisons of the strategies. All pairwise differences between Prompt A (Minimal), B (Structured), and C (Full-Context) are statistically significant (all $p < 0.001$), showing that increasing prompt complexity results in consistent, non-random improvements in quality of the reports. The large effect sizes ($r > 0.62$) sustain a strong and persistent improvement in clinical completeness throughout the entire 32-case set..

Comparison	W	p-value	Sig.	Effect r
A vs B	0	< 0.001	yes	0.879 (large)
B vs C	46	0.0004	yes	0.628 (large)
A vs C	0	< 0.001	yes	0.880 (large)

4.4 Faithfulness and Hallucination Analysis

Structured and full-context prompts resulted in perfect faithfulness (3.00 ± 0.00) with no hallucinated words on any of the 64 reports. Minimal prompting achieved 2.94 ± 0.35, corresponding to a 3.1% rate of hallucination. A coherent interpretation of these results is that the richer prompt context simultaneously increases both what the model is allowed to include (completeness) and constrain what it is allowed to fabricate (faithfulness). Importantly, Prompt A was the worst-performing prompt in both directions (lowest completeness and highest hallucination rate), indicating that what is missing from sparse prompts is partially filled in with plausible but ungrounded clinical details. This is consistent with robustness-related challenges in clinical LLM deployments [19,18].

4.5 Inter-Rater Reliability

Figure 3 shows the kappa results for each element. An overall $\kappa = 0.864$ (Almost Perfect) indicates that the completeness ratio yields fair and consistent results regardless of the score's strictness in or leniency.

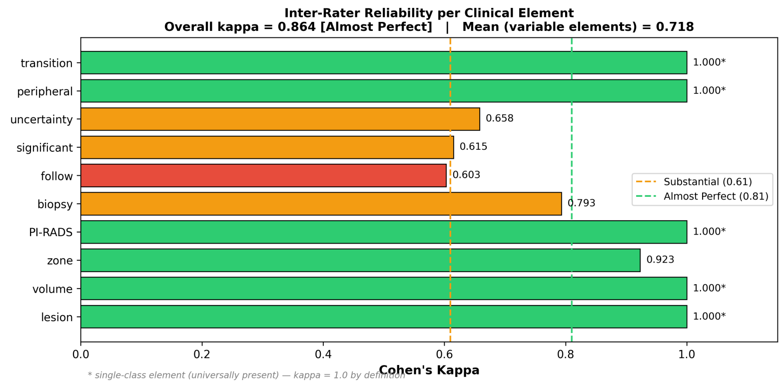

Fig. 3. Inter-rater reliability (Cohen's kappa) for each component of the clinical information between the conservative and the liberal scoring rules. Items denoted* are present in all instances—kappa = 1.0 by definition. Overall $=0.864\kappa =$ (Almost Perfect).

5 Discussion

5.1 Prompt Design is a First-Class Clinical Variable

Among the potentially surprising results of this study is that even the prompt design by itself can be a game changer. The completeness difference between Prompt A and Prompt C is an astonishing 48.7% points, Everything but the prompt was the same: the same model, the same data, the same infrastructure. This puts prompt engineering on a similar footing with architectural choices that impact the quality of clinical outputs. It also introduces a new dimension to what previous research found for medical question answering, where upgrading from a simple to a complex prompt did not consistently result in better outcomes [4]. For report generation, the connection is far clearer: any rise in the prompt complexity resulted in a larger effect size. As per the appendix on radiology prompting [6]: Nothing in rigor in prompt-design is non-negotiable — our data quantifies to what extent it is neglected: 48. 7 percentage points of omitted clinical content. The $W = 0$ effect size for A-vs-B and A-vs-C is worth dwelling on. In fact, not a single one of the 32 cases where the answer was wrong had that answer improved consensus by any tidier structured prompt. With effect sizes well beyond 0.62, these are not marginal improvements – they represent a strong, persistent uplift in report quality that is obvious throughout the data set.

5.2 Training-Free Accessibility

One practical attraction of LASRMIS is that it does not impose any requirement on the deployment side for training data, GPU resources, or model fine-tuning. You can get decent results just by writing a decent prompt and making an API call. This aligns with the philosophy of CBVLM [15], suggesting that conceptually meaningful concepts based training-free approaches can go head-to-head with trained methods – we bring that notion into the report generation space. Sophisticated methodologies that integrate Chain-of-Thought reasoning with retrieval augmented generation and fine tuning have also been demonstrated to improve clinical trustworthiness [16], and LASRMIS is not requesting that they be replaced. Instead, it is an instantiation of the intuition that a robust, prompt based instruction following LLM can get you very far, and that fine-tuning is something that you can afford to put off until you absolutely need near perfect completeness. This decoupling is crucial for low computing applications.

5.3 Clinical Safety Considerations

Prompt B and C did not hallucinate a single word on the 64 reports, so this looks good for the real world application. That said these are still draft outputs – each report must be checked by a radiologist prior to issuing for clinical use. LASRMIS is nothing more than the drawing board, not diagnosis, and nothing in the pipeline (at least as approved medical devices). Prompt A's 3.1% hallucination rate is a helpful reminder sparse prompts result in the model generating some of its own content, some of which will be clinically plausible, but factually unsupported in whole or in part. This is the precisely the type of behavior that metamorphic testing [19] and trust worthy AI frameworks [18] would question and why human oversight is an non-negotiable guardrail.

6 Limitations and Future Work

There are four major caveats to consider. The first experiments are based on 32 cases from a single clean benchmark; far from certain whether the pipeline can be applied to real PACS data where segmentation masks are imperfect or noisy. Second, the PI-RADS scores we assign are volume-based proxies - they are not true PI-RADS v2.1 assessments, which would necessitate T2 signal and diffusion analysis. Third, for reproducibility, we set temperature to zero and did not consider if more diverse results at higher temperatures would enhance the variety and naturalness of reports. The fourth is that our completeness and readability scores are automated proxies — they do correlate with clinical quality, but don't substitute for a radiologist's judgment, in particular since it's been demonstrated that LLM performance on radiology tasks appears to shift over time [14]. Earlier, the same three prompt strategies will be evaluated on GPT-4o [11] to verify if findings generalize across model families. Utilizing retrieval-augmented generation [16] may take hallucination rates even closer to the zero. Prospective validation with radiologist usability studies is a critical next step prior to any real deployment. Finally, the multimodal prompting explored for Alzheimer's disease reporting [17] points to a scheme for creating richer prompts that incorporate other image attributes that go beyond what segmentation masks can.

7 Conclusion

We created LASRMIS to test a simple but novel question: can off-the-shelf LLM, with a single prompt, translate prostate MRI segmentation metrics into clinically-styled radiology reports for the MR-only and MR-US workflows without training on either? Yes, and sensitivity to how to word the prompt; from 32 cases and 96 reports generated. Moving from the minimal to the full-context prompt increased clinical completeness by 48.7 percentage points (from 37.5% to 86.2%), with all pairwise comparisons being statistically significant and with large effect sizes. All conditions achieve high Faithfulness scores (2.94—3.00/3 to) and zero Hallucinations in Structured and Full-context Prompts. An inter-rater reliability of $\kappa = 0.864$ provides a good basis for us to confidently say that the rating scale captures something objective and stable. Overall, these results strongly suggest that prompt engineering is not a mere engineering detail—the quality of clinical output of a similar pipeline would be principally driven by it.

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